

OXIEDO: The Architecture That Unlocks Locked Data

The most valuable data in the world cannot be trained on.
ORMAS produces the account that
changes that, as the model learns it.

\$8.3B model-risk market • Obligations in law from Dec 2027 • Annual licence, on their infrastructure

| One Decision in 1986. Three Problems Everyone Funds Separately.

A Single Global Error Signal Makes Fault Localization Mathematically Impossible — Every Downstream Failure Follows From It

NOTHING CAN BE LOCATED

One error signal spreads across every parameter at once.

No unit can assess its own state.

The loss curve collapses and nothing says which node died.

NOTHING IS RETAINED

New tasks overwrite prior representations indiscriminately.

ResNet-18 keeps 47.3% after a sequential task shift.

Every regime change costs what the model knew.

NOTHING IS REPAIRED

Repair requires knowing what is broken.

Corrupted data is learned silently and confidently.

Rollback or full retrain are the only recourse.

One cause: the chain rule multiplies every unit's contribution through every other unit's. Unchanged since 1986.

Kill a Converged Layer. One Architecture Comes Back.

Both Networks Reach 85%. At Epoch 101 the Feature Layer Is Zeroed. Only One of Them Notices



CIFAR-10, parameter-matched baseline, three seeds. The stability result is local, not global, and has not been through peer review. Stated here rather than found later.

WHAT THE GAP IS

+70.3 pp

The baseline sits at chance **on every seed, permanently** — zero variance, no recovery.

ORMAS diagnosed the failure **inside one epoch** and climbed back through **85** individually attributed corrections.

Each node reads through a **4,416-parameter** bottleneck and fires at **2.5σ** below its own history.

| The Industry Is Moving From Inference to Training on Private Data

Three Independent Pressures Converge on the Same Missing Layer — None of Them Existed Two Years Ago

REGULATORS SHIFTED TO ARCHITECTURE

The FDA rejects medical models for opacity **regardless of accuracy**.

The EU AI Act attaches liability to systems with no audit trail.

The requirement is no longer performance. It is provenance.

RENTED INFERENCE STOPPED SCALING

Per-call cost compounds linearly with usage.

At institutional volume the only exit is your own infrastructure.

Which means training on your own data — the thing nobody can do safely.

MODELS DEGRADE INVISIBLY

Every regime change, patient cohort and fraud pattern erodes what the model knew.

No detection exists at the training layer.

It surfaces as a miscalculation, not an alert.

The regime is shifting from inference on public data to training on proprietary institutional data.

| The Market, Counted From the Constraint Upward

Sized Bottom-Up From Institutions That Qualify — Not Down From an Analyst Report

TAM \$8.3B (2026)

AI model-risk management, worldwide.

\$19.1B by 2035, 9.7% CAGR.

The budget exists. Today it buys tools that estimate a finished model from outside.

SAM \$250M

~1,000 institutions that qualify on all three tests:

- ~180 academic medical centres with an AI group and a regulated pathway
- ~400 funds and asset managers with in-house ML
- ~270 insurers and credit-risk teams under supervisory reporting
- ~150 labs running training at cost

× \$250K ACV = \$250M

SOM \$5M ARR

Beachhead: regulated finance.

Shortest cycle, clearest mandate, a named validator on the other side, and no ethics board.

20 institutions by end of year 3 at \$250K.

2% of SAM. The constraint is delivery capacity, not demand.

TAM: SNS Insider, AI model-risk management, 2026. Institution counts are our own estimates, and no buyer interviews have happened — the number most likely to be wrong.

| Who Signs, Who Uses It, and Who Can Kill the Deal.

One Beachhead Workflow — a Bank Whose Model Cannot Clear Validation

ECONOMIC BUYER

Head of Model Risk

Owens the MRM budget.

Wants: to clear validation first time.

Fears: being the name on a model an examiner reopens.

TECHNICAL USER

ML platform lead

Runs the training and owns the serving path.

Wants: to stop losing six-figure runs to late-found failures.

Fears: anything that changes inference cost.

RISK OWNER

Chief Risk Officer

Signs, and carries it to the board.

Wants: a defensible position, not a better number.

Fears: “the model changed and we cannot say how” in a supervisory letter.

GATEKEEPER

Independent validation

Internal team, then the examiner.

Wants: something verifiable rather than asserted.

Kills deals by: rejecting what it cannot reproduce.

The trigger is an event, not a budget cycle: a model fails validation or an examiner finding lands, and a funded programme cannot ship. **That is when this gets bought.**

No buyer interviews have been conducted. *This is the map the first twenty conversations are funded to correct.*

| One Product. Three Things Arrive With the Model.

Training Is Unchanged — Data, Objective and Resulting Model Are Identical. What Changes Is the Account That Exists Alongside It

01 — THE MODEL

Deploys exactly as it would have

Same architecture, same inference cost, no change to the serving path.

Nothing is traded away for the accountability. That is the whole point.

02 — THE NAMED PARTS

What it learned, per source

Each source's contribution held as a discrete addressable object, bound to the record of the data that justified it.

Removable on instruction, without retraining.

03 — THE RECORD

Every change training made

Which component, at which step, under which diagnosis, by how much, inside what declared limit.

Signed, and diffable against the last approved version.

Seven features of one architecture, not seven products. Nothing is priced per feature. anonymous.4open.science/r/ormas-EB73 — all 383 runs reproduce with one command.

| The Evidence Is Open. Replay a Real Run in Your Browser.

Checkable Without Us in the Room — and One of Them Is Running Live Right Now

LIVE oxiedo.com/black-box

An archived 54-day training run, replayable epoch by epoch, two identical networks side by side.

- Step through epochs 0–199 and watch per-component health move before the loss curve does.
- Open the correction ledger: step, component, diagnosis, magnitude, declared ceiling.

Seeds on screen. Every run regenerates on a customer's own hardware.

CONTROL A COMPANY, RUN AS AN EXPERIMENT

UK entity, real supplier, customers in **10 countries, 12 months, 7 pre-registered tests.**

Baseline \$6,691.68 across 79+ orders, **\$0** ad spend, **0** human sales hours.

Remove it **−91%** output, **−100%** new customers, all channels the same day, held 4 months.

Restore it **+1,300%**, overshoot **3.3×** on retained memory.

Ceiling test **\$4,386** single order, **\$0.00** CAC, no human in the transaction.

Tests written before the results, outcomes reported either way. This tested the orchestration layer — prior work, now closed. The architecture carries its own record: 383 controlled experiments, 4 families, one adverse result published alongside the rest.

| One Licence, Annual, Per Institution, On Their Metal.

Publish the Mechanism, License the Accountability — Priced Against Exposure Carried, Never Against Seats or Tokens

THE MODEL

Annual deployment licence, per institution, on-premise or in their own cloud.

No per-seat component. No consumption meter. No hosted tier, ever.

Target ACV **\$250K**, band \$150K–\$500K.

WHAT SETS THE FIGURE

Four inputs, written into the contract:

1. What the problem costs them today.
2. Model portfolio in scope.
3. Whether the bound exists or must be calibrated.
4. Liability scope accepted.

Assurance priced per engagement, on top.

WHY IT RENEWS

Artefact survival: on termination they keep every record, plus a frozen reader binary. Renewal is never hostage-taking.

Bound immutability: the declared bound is fixed at execution and moves only by signed amendment.

A validator who has accepted the method does not re-accept a replacement cheaply.

Gross margin is software-grade: nothing is hosted, so serving cost is the customer's. Zero contracts signed, zero customer conversations held — the first pricing conversation is the milestone, not the assumption.

| Go-To-Market: Three Motions, One Beachhead, Named Steps.

Regulated Finance First — Shortest Cycle, and a Named Validator on the Other Side

1 — TECHNICAL ENTRY

The falsification offer — under 30 GPU-hours, run by their own team.

Channel: named outbound to heads of model risk; the conference circuit.

Turns a sales call into an experiment.

2 — REGULATORY ENTRY

The validator is the real gatekeeper, so engage them before the customer does.

Channel: model-risk bodies, supervisory consultations, the Big Four validation practices.

A method a validator has seen is a shorter sale for every account after it.

3 — LAND AND EXPAND

Design partner (free, 3 months, read-only mode) → **one model in production** → **institution-wide licence**.

Read-only mode is the wedge: nobody lets an unproven system touch weights in a six-figure run.

Cycle: 6–18 months.

Y1 3 design partners, \$0, transformer result · **Y2** 4 licences, **\$1.0M ARR**, one validator acceptance · **Y3** 20 institutions, **\$5.0M ARR**, second sector.

Founder-led sales through years 1–2. The commercial hire lands against a pipeline, not ahead of one.

Named, Funded, and All Working One Layer Above the Constraint

Every Tool a Regulated Buyer Already Owns Inspects a Finished Model From Outside — None Observes Training From Inside

Named alternative	What it sells	Train-time record	Per-node attribution	Autonomous repair	Relevance to us
ModelOp	Governance control tower	×	×	×	4/5
Credo AI	Policy & compliance docs	×	×	×	3/5
Fiddler AI	Post-hoc explainability	×	~	×	3/5
Arthur AI	Production drift monitoring	×	×	×	2/5
Robust Intelligence (Cisco)	Pre-deployment stress testing	×	×	×	2/5
Retrain from scratch	The status quo remedy	×	×	×	5/5
ORMAS	The account, as it trains	✓	✓	✓	—

Complements, not casualties. All five inspect a model that has already made up its mind, so they share our buyer and budget line — **enterprise quotes from ~\$50K** — and none answers what training did. The real incumbent is the last row: **retraining from scratch, \$100K–\$500K a request.**

| One Founder Today. The Round Exists to Change That.

Key-Person Risk Is Total, Is the First Line of the Use of Funds, and Closes in This Order

WHAT IS ALREADY BUILT

Rokib Al Dhin Raadh — founder, relocating to San Francisco.

- The architecture, with a formal stability result, derived at 17.
- 383 experiments, reproducible by a customer's own team.
- Reviewer, **NeurIPS 2026** workshop committee — the field already treats him as a peer.
- A UK entity formed and run from abroad, unassisted.

THE HIRING SEQUENCE

Month 1 Research engineer.

Removes single-point failure on the architecture itself.

Month 9–12 Commercial lead who has sold inside a fund, hospital or insurer. Lands against a pipeline, not ahead of one.

Month 12–18 Assurance contractor.

Eight people by end of year 3.

WHY YEAR 1 DOES NOT NEED A SALESPERSON

Y1 is a **technical evaluation**, not procurement: a 30 GPU-hour run a head of model risk authorises without a purchase order.

Founder-led is correct here: the first conversations are with people who want to attack the method.

No licence is sold before the commercial lead is in place.

The honest version: never having worked inside a team is the gap, and the sequence above is how it closes. raadh.me · oxiedo.com

Financials: Burn, Runway, and the Unit Economics Behind Them.

Pre-Revenue by Design — Nothing Ships Until the Architecture Clears Its Own Destruction Suite

	Year 1	Year 2	Year 3	Assumption
Paying institutions	0	4	20	Land at 1 model, then expand
ARR	\$0	\$1.0M	\$5.0M	\$250K ACV, held flat
Gross margin	—	~92%	~92%	On-premise; no hosting cost
CAC (fully loaded)	—	~85K	~85K	Founder/commercial time, pilot eng.
LTV (4-yr, GM-adj.)	—	\$920K	\$920K	4-year life, no expansion assumed
LTV:CAC	—	~10.8:1	~10.8:1	Benchmark 3:1 healthy, 5:1 top
Net burn	\$820K	\$680K	Near breakeven	Compute front-loaded to Y1

Every number is an assumption, and the ratio flatters because the inputs are unproven. \$250K ACV is benchmarked against model-risk tooling and validated by nobody — zero pricing conversations have happened. Y2 assumes the transformer result lands and one validator accepts the method; if either fails, Y2 is 1-2 design partners and a seed raised on research.

Seed assumed at month 15 against milestones, not ARR. This round does not reach default-alive and is not shown as though it does.

| From a Training Layer to the Substrate Regulated AI Runs On

Every Phase Removes One Dependency — Ending With a Stack That Rents Intelligence From No One

PHASE 1 — NOW

Finish the research and carry per-node health monitoring onto attention heads.

Architecture-agnostic by design, so this is a compute problem, not a research one.

First design partners.

PHASE 2 — 12-24 MONTHS

First licensed deployments inside institutions that cannot move their data.

Each accumulates institutional memory that cannot be exported.

Switching cost compounds with tenure.

PHASE 3 — YEARS 3-5

A model pre-trained on the self-correcting architecture from scratch.

The external dependency disappears; cost collapses to hardware.

A position no competitor reaches by adding a feature.

The network learns to stop needing correction. **That is the thesis, and it is measured.**

Correction frequency decays $4.2 \rightarrow 0.05$ per epoch — the system repairs itself until it no longer needs to.

| \$1.5M Pre-Seed on a SAFE. Sized to the Next Result.

Eighteen Months, Six Milestones With Acceptance Criteria — the Compute Line Costed Against the Question Actually Open

THE INSTRUMENT

\$1.5M on a post-money SAFE.

\$15M post-money cap, no discount, no debt.

Dilution at conversion: **10.0%**, fixed on signature.

Cap-only is how 72% of SAFEs are written. Priced round deferred to the seed.

USE OF FUNDS

\$650K Research engineer; co-founder, equity-weighted

\$250K 32–64 accelerators, two campaigns

\$150K One custodian, non-exclusive

\$100K Delaware, EU rep, filings

\$200K 18 months of everything else

\$150K Buffer, unallocated

WHAT IT BUYS

1. A second and third person. Key-person risk is total today.

2. One result that is not CIFAR. Bounded attribution on a transformer backbone.

3. Three design partners, and this company's first pricing conversation.

The frontier campaign is a seed problem, not this one.

Before writing a cheque, spend \$50 instead: the remaining programme is falsifiable in under 30 GPU-hours, on your hardware, without us in the room.

APPENDIX

The questions that come next

A1 What is not established · A2 The falsification specification · A3 Where the numbers come from

| A1 — What Is Not Established

The Boundaries of the Evidence, in the Same Place as the Evidence

THE LIMITS

- Every result is **CIFAR-10 or CIFAR-100**. No clinical, financial, biological or defence data has touched the system.
- Demonstrated on fully-connected, convolutional and residual families. **Transformers are a target, not a result.**
- Test-time distribution shift is unrun: the health signal comes from gradients, and at inference there are none.
- Whether the signal survives at small n is untested.

THE ADVERSE RESULT

Under **adversarial weight injection** the architecture measures **1.0 pp worse** than a parameter-matched baseline.

Adversarially crafted perturbations hold nominal activation statistics while moving decision boundaries — which evades precisely what this monitors.

Published rather than omitted, and unmitigated.

It is on the technology page and named in every market where it bears on a decision.

The claim is transparency, not state-of-the-art accuracy. The accuracy gap is the evidence for it.

A2 — Falsify This for Under \$50

Four Measurements Decide the Rest of the Programme, and Together They Cost Under Thirty GPU-Hours

Measurement	What a negative result would mean	Cost
Archive analysis	The weak form of the central question, against 67 archived runs.	Zero GPU
Monitoring lead time	How early per-component health crosses threshold. Opens or closes a market.	8.4 GPU-hr
Wrapper viability	Whether a model we did not train keeps enough component independence.	~1 GPU-day
Separation study	Whether component disagreement separates mislabelled from merely hard data.	~20 GPU-hr

None has been run. A negative result narrows the programme rather than ending it — the shipping features depend on none of the four. **Run it on your hardware, without us in the room.**

A3 — Where the Numbers Come From

Every Figure in This Deck, With Its Source and Whether Anybody Has Checked It

Figure	Source	Status
80.3% vs 10.0%, +70.3 pp	383 controlled experiments, reproducible from seed	Measured, replayable
−91% / +1,300%	12-month ablation, Black Bloxie LTD, SHA-256 reconciled	Measured, prior work
\$8.3B TAM, 9.7% CAGR ~1,000 institutions	SNS Insider, AI model-risk management, 2026 Our own bottom-up count across four segments	Third-party analyst Estimate, unvalidated
\$250K ACV, \$150K–\$500K CAC \$85K, LTV:CAC 10.8:1 Competitor pricing ~\$50K \$15M cap, 10% dilution	Benchmarked against model-risk tooling Modelled on a 6–18 month enterprise cycle Published enterprise quote floors 2026 AI pre-seed band \$12–25M; \$15M is the norm at this raise size	Zero pricing conversations Assumption, no data Third-party Market benchmark

Four of these seven are unvalidated, and they are the four the round exists to validate. An investor who takes the top two and discounts the bottom four is reading this deck exactly as intended.